

## SYDNEY CAMDEN, Australia

WMO: 947550

Lat: 34.0392S

Lon: 150.6889E

Elev: 245

StdP: 14.57

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 32.2       | 34.4 | 26.7                          | 20.8 | 58.3 | 29.0 | 23.2 | 54.5 | 24.5                  | 59.1 | 21.3 | 59.3 | 1.3                   | 150  | 0.400 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 20.9                   | 95.9            | 68.8 | 91.5 | 68.5 | 87.4 | 67.5 | 73.3                | 85.2 | 71.8 | 82.4 | 70.7 | 80.6 | 11.2                 | 10   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.6                            | 113.6 | 75.5 | 69.2 | 108.2 | 74.5 | 67.8 | 103.2 | 73.9 | 37.0          | 85.1 | 35.8 | 82.2 | 34.7 | 80.7 | 80.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 20.7          | 17.7 | 15.3 | DB                         | 28.2 | 104.5              | 1.8 | 4.2                                                | 26.9 | 107.5      | 25.8 | 110.0      | 24.8 | 112.3      | 23.5 | 115.3 | (4) |
| (5)               |      |      | WB                         | 27.8 | 76.5               | 2.0 | 1.8                                                | 26.4 | 77.8       | 25.2 | 78.8       | 24.1 | 79.8       | 22.7 | 81.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 62.4        | 73.9  | 72.7  | 69.4  | 63.3  | 56.4  | 52.1  | 50.5  | 52.4  | 58.0  | 62.8  | 67.1  | 70.9  | (6)  |  |
|                                                                        | DBStd       | 9.35        | 5.41  | 5.05  | 4.47  | 4.41  | 4.14  | 3.75  | 3.53  | 3.61  | 4.64  | 5.35  | 5.81  | 5.47  | (7)  |  |
|                                                                        | HDD50       | 76          | 0     | 0     | 0     | 0     | 3     | 21    | 36    | 15    | 1     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 1970        | 1     | 2     | 11    | 83    | 267   | 387   | 451   | 390   | 217   | 109   | 42    | 10    | (9)  |  |
|                                                                        | CDD50       | 4600        | 741   | 635   | 602   | 398   | 202   | 84    | 50    | 91    | 241   | 396   | 512   | 648   | (10) |  |
|                                                                        | CDD65       | 1017        | 277   | 217   | 148   | 31    | 1     | 0     | 0     | 0     | 6     | 40    | 104   | 193   | (11) |  |
|                                                                        | CDH74       | 8586        | 2235  | 1585  | 935   | 280   | 21    | 0     | 2     | 9     | 186   | 621   | 1052  | 1660  | (12) |  |
|                                                                        | CDH80       | 3370        | 1000  | 627   | 279   | 61    | 0     | 0     | 0     | 0     | 43    | 221   | 438   | 701   | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.5         | 6.1   | 5.7   | 5.1   | 4.4   | 4.5   | 4.6   | 5.1   | 6.2   | 6.2   | 6.1   | 6.3   | 6.2   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 33.7        | 3.9   | 4.1   | 3.9   | 3.2   | 2.3   | 2.6   | 1.6   | 1.3   | 1.6   | 2.8   | 3.0   | 2.1   | (15) |  |
|                                                                        | PrecMax     | 49.6        | 10.4  | 16.6  | 13.5  | 13.3  | 6.1   | 12.0  | 5.6   | 6.5   | 5.8   | 9.6   | 6.5   | 6.4   | (16) |  |
|                                                                        | PrecMin     | 17.6        | 0.4   | 0.6   | 0.2   | 0.1   | 0.1   | 0.2   | 0.0   | 0.0   | 0.1   | 0.0   | 0.4   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 9.2         | 2.8   | 4.1   | 3.2   | 3.3   | 1.8   | 2.8   | 1.5   | 1.9   | 1.4   | 2.5   | 1.8   | 1.7   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 102.2 | 99.8  | 92.3  | 86.9  | 76.7  | 69.5  | 71.0  | 74.9  | 85.3  | 92.5  | 98.4  | 99.8  | (19) |  |
|                                                                        |             | MCWB        | 71.9  | 69.9  | 67.9  | 64.0  | 59.1  | 56.3  | 55.4  | 55.9  | 59.8  | 62.7  | 64.8  | 68.7  | (20) |  |
|                                                                        | 2%          | DB          | 95.2  | 92.6  | 86.5  | 80.4  | 72.9  | 66.1  | 66.1  | 70.4  | 79.2  | 86.2  | 90.5  | 93.0  | (21) |  |
|                                                                        |             | MCWB        | 71.0  | 70.5  | 68.3  | 63.4  | 58.1  | 55.0  | 52.4  | 54.5  | 58.2  | 62.3  | 66.6  | 67.8  | (22) |  |
|                                                                        | 5%          | DB          | 90.1  | 87.0  | 82.6  | 76.8  | 70.0  | 63.9  | 63.5  | 67.0  | 74.7  | 81.1  | 84.4  | 87.3  | (23) |  |
|                                                                        |             | MCWB        | 70.2  | 69.5  | 67.3  | 62.6  | 57.4  | 53.7  | 51.6  | 53.0  | 57.7  | 61.5  | 64.8  | 66.8  | (24) |  |
|                                                                        | 10%         | DB          | 85.4  | 82.9  | 79.1  | 73.5  | 67.4  | 61.8  | 61.2  | 64.0  | 70.6  | 76.1  | 79.7  | 83.0  | (25) |  |
|                                                                        |             | MCWB        | 68.9  | 68.5  | 66.1  | 61.6  | 56.3  | 53.0  | 50.8  | 51.7  | 56.1  | 60.1  | 63.9  | 66.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 75.8  | 75.6  | 72.7  | 68.4  | 64.6  | 59.9  | 58.9  | 59.0  | 63.8  | 67.8  | 71.6  | 73.5  | (27) |  |
|                                                                        |             | MCDB        | 91.0  | 88.0  | 81.0  | 78.5  | 68.2  | 63.6  | 63.4  | 68.0  | 73.9  | 79.2  | 85.5  | 87.2  | (28) |  |
|                                                                        | 2%          | WB          | 73.5  | 73.2  | 70.9  | 66.3  | 61.8  | 57.7  | 55.8  | 56.8  | 61.7  | 65.4  | 69.1  | 71.3  | (29) |  |
|                                                                        |             | MCDB        | 86.7  | 83.8  | 78.9  | 73.3  | 67.0  | 61.8  | 60.5  | 65.2  | 71.6  | 76.8  | 82.0  | 84.0  | (30) |  |
|                                                                        | 5%          | WB          | 71.9  | 71.7  | 69.5  | 64.9  | 59.9  | 56.3  | 54.0  | 55.0  | 60.0  | 63.7  | 67.3  | 69.7  | (31) |  |
|                                                                        |             | MCDB        | 83.2  | 80.7  | 77.4  | 71.8  | 66.3  | 60.8  | 59.6  | 63.3  | 69.4  | 75.1  | 78.9  | 81.6  | (32) |  |
|                                                                        | 10%         | WB          | 70.5  | 70.4  | 68.2  | 63.5  | 58.2  | 54.9  | 52.4  | 53.3  | 58.2  | 62.1  | 65.7  | 68.2  | (33) |  |
|                                                                        |             | MCDB        | 80.7  | 78.8  | 75.8  | 70.4  | 64.8  | 59.7  | 58.8  | 61.4  | 67.0  | 72.6  | 75.7  | 78.5  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 20.9  | 19.3  | 18.9  | 21.3  | 23.4  | 21.1  | 23.7  | 25.1  | 25.2  | 24.1  | 21.8  | 21.5  | (35) |  |
|                                                                        |             | MCDBR       | 31.2  | 28.2  | 25.9  | 27.2  | 27.8  | 25.0  | 28.4  | 31.3  | 34.0  | 35.3  | 33.1  | 31.2  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.5   | 8.8   | 9.5   | 11.7  | 14.2  | 14.6  | 16.1  | 16.2  | 14.7  | 13.6  | 11.5  | 10.3  | (37) |  |
|                                                                        |             | MCWBR       | 24.9  | 21.9  | 20.7  | 21.4  | 21.3  | 19.0  | 21.4  | 25.9  | 26.8  | 26.8  | 25.8  | 25.5  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.397 | 0.392 | 0.362 | 0.334 | 0.316 | 0.305 | 0.296 | 0.304 | 0.337 | 0.358 | 0.385 | 0.400 | (40) |  |
|                                                                        |             | taud        | 2.420 | 2.456 | 2.515 | 2.566 | 2.567 | 2.596 | 2.602 | 2.573 | 2.481 | 2.452 | 2.419 | 2.386 | (41) |  |
|                                                                        | Ebn at Noon |             | 298   | 293   | 291   | 282   | 271   | 266   | 275   | 288   | 294   | 300   | 300   | 298   | (42) |  |
|                                                                        |             | Edh at Noon | 39    | 36    | 32    | 27    | 24    | 22    | 23    | 26    | 32    | 36    | 39    | 41    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2033  | 1726  | 1480  | 1207  | 962   | 757   | 886   | 1171  | 1561  | 1807  | 1908  | 2090  | (44) |  |
|                                                                        | RadStd      |             | 160   | 162   | 133   | 105   | 64    | 67    | 68    | 72    | 95    | 171   | 178   | 193   | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |  |
| (46) | Station Only            | +0.88 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -131  | +373  | +211  | (46) |  |
| (47) | Regional (37 neighbors) | +0.79 | N/A     | N/A    | +1.39   | N/A   | N/A   | -4          | -149  | +328  | +169  | (47) |  |

Nomenclature: See separate page